

We thank you for your time spent taking this survey.
Your response has been recorded.

Research Ethics Application Form – CEP

Principle Investigator and Student Details (to be completed by all students)

Section 1a: Details of Principal Investigator (PI)

For all projects the PI must be employed by Imperial College London or hold an honorary contract.

For all student projects, the student's supervisor must be named as PI.

Student, co-investigator and collaborator details must be added to Section 13.

Supervisor (PI) name (incl. title)

Dr. Iain Staffell

Academic position (at Imperial College London)

Associate Professor in Sustainable Energy

Faculty

Faculty of Natural Sciences

Division/ School/ Department

Centre for Environmental Policy

PI Email (Imperial College)

i.staffell@imperial.ac.uk

Summary of PI skills (experience relevant to the study and in any procedures to be used) (350 characters max)

Dr. Iain Staffell is Associate Professor of Sustainable Energy at Imperial's Centre for Environmental Policy, with 15 years' experience in energy research. He co-developed Renewables.ninja and related tools, leads the Electric Insights project, and is widely cited on energy and decarbonisation in international media.

Section 1b: Student Details

Student name

Putera Hendri Riyanto

Student email (Imperial College)

putera.riyanto25@imperial.ac.uk

Section 2: Research Type

Is your study considered low risk?

'Low Risk' includes collecting data from individuals or groups that can give consent, or using secondary data that you have permission to use.

See the Research Term Guidance document for more information and consult with your supervisor.

☒ **Yes**

☐ No

Does your study only involve analysis of secondary data which is publicly available, and permission is not required to access the data? This includes extended or systematic literature reviews.

☒ **Yes**

☐ No

Section 3: Filter for ICREC and SETREC

Is the primary aim of the research answering a human health related question?

☐ Yes

☒ **No**

Is the primary aim of the research answering a non-health related science, social science, engineering or technology related question?

☒ **Yes**

☐ No

Is the primary aim of the research to answer an educational question?

☐ Yes

☒ **No**

Section 4: Risk level categorisation

This section determines the research risk level and if the application requires full committee review.

Meeting dates and submission deadlines

[ICREC](#)/[SETREC](#).

High risk

Yes

No

Does the research involve drugs/medication? If yes, please attach the SmPc.



Does the research involve genetically modified materials? If yes, please also complete appendix two and attach the GM Safety Committee letter.



Will you be recruiting vulnerable participants? i.e. children (15 years or younger), adults (16 years or over) who are unable to consent, people in care, the mentally ill or individuals with learning difficulties?



Will participants take part in the study without their explicit consent? i.e. studies involving deception.



Will you be recruiting prisoners or young offenders?



Yes

No

Is there any aspect of the proposed research which could potentially cause harm to the reputation of the College? i.e. could the research be considered controversial or prejudiced?

☐☒

Could participants disclose any illegal or harmful activity due to the nature of the research?

☐☒

Will personally sensitive subjects be discussed that have the potential to induce stress, anxiety or negative consequences for the participant?

☐☒

Will the researcher be in a position of influence or authority over the participants that could give rise to a perceived pressure to participate? i.e. lecturers/teachers and students.

☐☒

Does the study involve physically intrusive procedures, administration of substances, use of bodily fluids, tissues, DNA or RNA? Use of relevant material must be registered with Imperial College Tissue Bank under the College HTA license.

☐☒

Yes

No

Does the study involve ultrasound or sources of non-ionizing radiation? i.e. radiation, MRI, or fMRI.

☐☒

Are there any potential conflicts of interest, or what could be perceived by an outside observer as conflicts of interest?

☐☒

Will undue incentives for participants be offered? Incentives should be proportionate to the burden imposed and justified by the benefits.

☐☒

Are you using any medical device in the UK that is CE/UKCA marked but is being used outside its product limitation? Or are you using any non-CE/non-UKCA marked product(s)? For more information on [regulating medical devices](#).

☐☒

Does the proposed research raise any ethical issues that are not covered above?

☐☒

If you have answered NO TO ALL of the above questions, your project is considered low risk and you can proceed with the rest of the ethics form.

Section 5: Project Description

(to be completed by all)

Full title of study

Indonesia's Climate Menu: Adapting the Global Climate Stabilisation Wedges Framework to a National Scale

P code or cost code and study funder
(only if applicable to study)

N/A

Lead organisation
(who has overall responsibility for the study)

Imperial College London

List of location(s) where study will be conducted

London, United Kingdom

Proposed start date

From start of advertising and/or recruitment

May 18th 2026

Proposed end date

To end of data collection

September 2nd 2026

Section 6: Project Summary

Provide a summary of the project in **lay terms**, this should include:

- a brief description of reasons for doing the study (justification for the research)
- the aims (research aim/question)
- how data will be disseminated
- any expected benefits to the participant, researchers or others.

(500 words max)

Indonesia is one of the world's top carbon emitters, with emissions primarily originating from agriculture, forestry, and other land-use activities (AFOLU) and energy. Under its Long-Term Strategy for Low Carbon and Climate Resilience (LTS-LCCR), Indonesia commits to implementing a scenario aligned with the Paris Agreement, with an emissions target of 540 MtCO₂e by 2050 compared with 2,454 MtCO₂e under current policies. To implement the low-carbon scenario, Indonesia plans to focus heavily on the forestry sector, followed by the energy sector (increasing renewables and efficiency) and Carbon Capture and Storage (Ministry for Environment and Forestry of Indonesia, 2021). Given the potential trade-offs across sectors and the ambitious target to be met, Indonesia could use a framework that accounts for country-specific technological, economic, and socio-political factors to help its stakeholders explore and compare mitigation strategies. In a global context, Pacala and Socolow (2004) introduced the concept of a stabilisation triangle, which segments the carbon emissions trajectory into wedges. They argue that carbon emissions should be limited to 7 GtC annually over the subsequent 50 years, whereas levels could reach 14 GtC annually under a Business-as-Usual (BAU) scenario. Within this framework, a wedge is defined as a strategy to reduce carbon emissions by 1 GtC annually by 2054; therefore, 7 wedges are needed, with 15 potential strategies identified. Johnson and Staffell (2026) expanded the framework by considering the trajectory of global greenhouse gas (GHG) emissions, updated climate goals, and emerging mitigation strategies. As a result, the wedge definition was changed to 2 GtCO₂e/year (0.54 GtC/year), the timeframe was shortened to 30 years, and the number of strategies

increased to 36. Despite these developments, no study has yet applied this global climate stabilisation wedges framework at the national level. This leaves a gap in accessible, country-specific decarbonisation tools, particularly given that national-scale governance has greater authority in formulating climate policy for its country (Johnson & Staffell, 2026). This project aims to develop a national-scale climate stabilisation wedges framework for Indonesia by quantifying the mitigation potential and deployment scale of each strategy using the global climate stabilisation wedges framework to meet Indonesia's 2050 climate target. Objectives in achieving the aims are as follows: a. Collect and analyse Indonesia-specific data from national and international sources to underpin the national stabilisation wedges framework. b. Define and justify the size of a wedge for Indonesia for rescaling the global climate stabilisation wedges to Indonesia's context. c. Estimate the number of wedges required to achieve Indonesia's 2050 climate target. d. Reassess the effort (deployment scale) and upper bound (potential) of each strategy within Indonesia's context. The project will assess how Indonesia's decarbonisation pathways differ from the global framework, helping stakeholders in Indonesia evaluate and explore pathways relevant to Indonesia's context.

Section 7: Research Methods

What methods will you be using in this study?

Briefly describe in **lay terms**:

- what will happen
- the number of times
- any data collection techniques

(500 words max)

The project will use publicly available national and international data and use it within an existing framework to develop a new one. By quantifying each strategy's mitigation potential and implementation scale in Indonesia, the project will assess how Indonesia's decarbonisation pathways differ from the global framework. A quantitative, secondary data analysis approach is appropriate given the range of accessible databases available through Imperial College London's library, and it suits projects that examine spatial differences, such as this one (Davies & Hughes, 2014). This project will need sequential steps: a. Review core and related literature, including the calculation method underpinning the global wedge framework in the supplementary materials of Johnson & Staffell (2026), to understand the methodology before adapting it to the national scale. b. Collect Indonesia-specific data for strategies provided from the global framework, both from national sources, international sources, and peer-reviewed literature. Datasets might be inconsistent, incomplete, or not sufficiently detailed for the project's context. Additionally, national and international data may differ in their boundaries, scope, units, timeframes, and methods, making direct comparison difficult. To mitigate this, the project will prioritise available national government data, as it should be the most relevant source reflecting conditions in Indonesia, and use datasets from international bodies and peer-reviewed literature as fallbacks to fill gaps. Should the data differ across sources or be unavailable, transparent assumptions will be presented to justify their use in this project. c. Define a wedge in Indonesia's context. The project will first identify the Business-as-Usual and decarbonisation scenarios for Indonesia in 2050 to determine the emission gap to be reduced. Then, the project will use two principles (population- and GDP-based) to compare the size of one Indonesian wedge under each, ultimately justifying the most suitable wedge size for Indonesia's context. d. Recalculating effort and upper bound for each strategy. The project will use Indonesia-specific data within the existing global framework, using the selected wedge size from the previous step. In identifying the deployment scale required to achieve

one wedge and the upper bound for each strategy in Indonesia's context, this project will provide feasible comparisons and transparent assumptions when data are inconsistent or limited.

STUDY OBJECTIVES

What are you hoping the study achieves? List the primary, secondary and other study objectives.

List the specific research objectives. Again, this can be copied from your research proposal

The aim of this study is to develop a national-scale climate stabilisation wedges framework for Indonesia, adapted from global climate stabilisation wedges framework, by quantifying the mitigation potential and deployment scale of each strategy to meet Indonesia's 2050 climate target. The objectives in achieving this are:

- a. Collect and analyse Indonesia-specific data from national and international sources to underpin the national stabilisation wedges framework.
- b. Define and justify the size of a wedge for Indonesia for rescaling the global climate stabilisation wedges to Indonesia's context.
- c. Estimate the number of wedges required to achieve Indonesia's 2050 climate target.
- d. Reassess the effort (deployment scale) and upper bound (potential) of each strategy within Indonesia's context.

PUBLICATION POLICY

The study publication policy should be described in full.

In most cases, you should just state that this work is being prepared for publication as an MSc thesis at Imperial College London. If you intend to write a report for some external party or you intend to publish in the academic/peer reviewed literature you should say that too

This work is being prepared for publication as an MSc thesis at Imperial College London.

STUDY DESIGN

Detail how this study will be conducted from advertising to data processing. Make sure your design is explicitly described and appropriately justified. **Repeat for each method you use.**

Include the following bullet points below:

- Type of study method: survey, interviews, focus groups, simulator study.
- Recruitment methods: how you will recruit participants, use of participant information sheets and consent forms
- Duration of study from advertising and recruitment to end of data collection
- Number of participants. This should be justified.

This should describe what variables you will be measuring, how you will be measuring them, how often they will be measured (just once, or is there a follow-up or repeat measurement) and whether this will be done through survey, interview, focus group or other means. Describe here how many people you are looking to recruit and how you will recruit them. Any further details about inclusion criteria will be given in the next sections.

As this project will use publicly available secondary data, no study methods, such as surveys, interviews, or focus groups, will be conducted. Alternatively, the project will construct an initial data inventory after determining the required data types using the Global Climate Stabilisation Wedges framework. Then, a comprehensive data inventory will be compiled following thorough searches of national and international datasets and the peer-reviewed literature. These data are used to define the new size of a wedge in Indonesia using the most appropriate principle and reassess the effort (deployment scale) and upper bound (potential) of each strategy within Indonesia's context. The effort reassessment and upper-bound estimates will be iterated based on the number of strategies that are well-suited to Indonesia's technological, economic, and socio-political context.

Section 8: Participant Recruitment

Provide details of methods of recruitment, participant inclusion and exclusion criteria and the number of participants you are aiming to recruit.

Please include:

- Details of any incentives (such as financial reimbursement).
- Recruitment and advertising material (email, poster, social media advert)
- Oral information scripts

*The recruitment material (points 2 and 3) needs to be developed in a separate document and sent to your supervisor along with the print out of your ethics form.

(1000 words max)

There will be no Participant Recruitment

Pre-recruitment evaluations

Are there any requirements that a participant must fulfil? E.g., fitness test, allergy test.

This will likely be not applicable to most of you – you should just say that there will be no pre-recruitment evaluations

There will be no pre-recruitment evaluations

Inclusion criteria

Include justifications.

This is relevant if you are, for instance, interviewing professionals or members of a particular group. Indicate why you need to speak with these people specifically and how you will know that they meet your specific criteria. For instance, “I will contact the sustainability officers of large multinational companies based on their publicly available LinkedIn profiles. The study requires insights from professionals on [whatever the topic is...]”

There will be no Participant Recruitment

Exclusion Criteria

Include justifications.

If there is a reason why you might exclude someone after you contacted them, you should say this here. Otherwise, you may just reiterate that anyone meeting the inclusion criteria will be considered and there are no specific exclusion criteria. Extending the above example you might be looking for experts, so one of your exclusion criteria might be that if you have identified someone through LinkedIn as a sustainability officer, but it turns out they have less than 1 year experience in the role, you will exclude them

There will be no Participant Recruitment

Section 9: Informed Consent

Include details of how you will be obtaining consent.

Detail the process for ensuring informed consent of all research participants.

As there will be no Participant Recruitment, informed consent will not be required.

The withdrawal process(es).

Describe procedures for

- stopping early
- when can a participant withdraw from the study
- how and who to contact.

Include details of data usage once participant has withdrawn if the data be used once it has been anonymised.

As there will be no Participant Recruitment, informed consent will not be required.

If participants whose first language is not English are to be recruited, state clearly how the details of the study will be explained, and the consent processed.

As there will be no Participant Recruitment, informed consent will not be required.

Section 10: Ethical Summary

Has any part of this proposal received prior ethics approval?
e.g. if project part of a larger project that has already gone through ethics, then an amendment to that ethics approval will need to be sought via College committees – may be lengthy – avoid if possible, unless minor.

☐ Yes

☒ **No**

Is this study subject to local ethics approval to where research is being conducted?
e.g. through another university or institution

☐ Yes

☒ **No**

Section 11: Documentation checklist

Do Imperial College's insurers need to be notified about your project?

If your project is running abroad and is not qualitative or data only, or if your project is interventional and involves pregnant women, children under 5 or more than 5000 participants you may need additional insurance cover. [Insurance for studies](#), email the [insurance team](#) with any insurance enquiries.

If yes, please provide confirmation that insurance cover has been agreed.

☐ Yes

☒ **No**

☐ In Process

Has your research project been independently peer reviewed?

This can be organised by the [Peer Review Office](#) (within the RGIT). If you answered yes to any questions in section 3, you may be asked to ensure the study is peer reviewed. However, the study does not have to use the RGIT's office for peer review.

☐ Yes

☒ **No**

☐ In Process

Are you developing a mobile app?

See the [mobile app webpage](#) for more information.

☐ Yes

☒ **No**

☐ In Process

Have you had a Disclosure and Barring Service (DBS) check carried out? If yes, when (add date).

For more information about DBS, check [government guidance](#) and the [College website](#).

☐ Yes

☒ **No**

☐ In Process

Do you need a contractual agreement in place? For further information, please contact your supervisor in the first instance.

☐ Yes

☒ **No**

☐ In Process

Do you have permissions to use the data in your study? This may be required if you are looking at secondary data.

Postings on social media may not be considered public for research uses – please check

- ☐ Yes
- ☐ No
- ☐ In Process
- ☒ **Public data, permission not required**

Has Imperial College's [Risk Assessment procedure](#) been followed? Contact your departmental administrator for further information.

- ☒ **Yes**
- ☐ No
- ☐ In Process

Section 12: Confidentiality and management of personal and other research data

I understand it is the responsibility of the researcher to ensure all research data is securely stored during and after the study in accordance with College Guidelines, Codes of Practice, Policies and Procedures.

☒ **Yes**

I confirm that all the processing of personal information related to the study will be in full compliance with the GDPR. Including but not limited to, the creation of all necessary documentation (PIS, Data Protection Impact Assessments, Consent forms etc.)

☒ **Yes**

Describe how you will preserve the confidentiality of participants taking part in the study and fulfil transparency requirements under the General Data Protection Regulation.

Best practice requires that all participants information be entirely anonymised. If you are not doing this please give well-justified reasons for keeping personal and traceable information.

All data should be stored on a GDPR-compliant service e.g. Imperial College OneDrive or Sharepoint.

The project will use only publicly available secondary data from national governments and international institutions (e.g. national GHG inventories, national statistics, IEA country data, FAOSTAT, Climate Watch, Indonesia's NDC, and LTS-LCCR 2050). Hence, there should be no major issue regarding confidentiality. Nevertheless, as the data may be inconsistent and imperfect, this project will make its assumptions transparent to justify its use of the data and prevent misrepresentation. GDPR and Imperial's guidelines will be followed for data processing and storage.

Section 13: Co-investigators/ Collaborators (second supervisor or external collaborators)

(to be completed by all)

If there are more than four co-investigators (second supervisor or external collaborators), please use a separate sheet and follow the format below.

Name of Co-investigator/ Collaborator

Dr. Nathan Johnson

Position Incl. organisation, company, institution

Research Associate, Centre for Environmental Policy, Faculty of Natural Sciences,
Imperial College London

Role in the study (what contributions you will make and relevant experience)

2nd Supervisor. Dr. Nathan Johnson is a multi-disciplinary scientist working in the fields of energy, land use and climate change. Nathan's research centres around modelling solutions for decarbonising the global economy and mitigating climate change. Nathan's PhD applied the well-known Stabilization Wedge framework to estimate the climate impact of a diverse set of technological, behavioural, and natural climate solutions.

Email (Work, not personal)

nathan.johnson17@imperial.ac.uk

Do you have another Co-investigator/ Collaborator (second supervisor or external collaborators)?

☐ Yes

☒ **No**

PI Declaration

I declare that:

- I undertake to abide by the ethical principles underlying the Declaration of Helsinki (1964) and subsequent amendments and good practice guidelines on the proper conduct of research.
- I undertake to abide by the Data Protection Act 2018 and General Data Protection Regulation (Europe) and any applicable local laws.
- I undertake to abide by all local laws and regulations for non-UK research.
- I will report any adverse or unforeseen events or protocol violations and deviations which occur to the Ethics and Research Governance Co-ordinator within 24 hours.
- I will provide an [annual progress report](#) of the project until the end of the study.
- If I register my study on a public database, i.e. ClinicalTrials.gov, I will report results on that database within one year of study completion.
- I will provide [notification of the end or early termination](#) of the research project.
- I will provide [notification of amendment](#) to the CEP Ethics Panel if there are any changes to the research protocol or personnel which affect the ethical aspects of the project.
- I will assist the CEP Ethics Panel in any continuing review of the project deemed necessary by the Committee or Faculty Members.

- All information on this form is correct.

PI Name (please print)

Dr. Iain Staffell

PI Signature: PI must sign and date this section after you have submitted and downloaded the PDF form

(Note: PDF copy of form can be downloaded on final page of survey following submission)

✕  RE

clear

Date

May 22th 2026

Any attendance required by the CEP Ethics Panel must be made by the PI named in section one. Attendance at the meeting will give you the opportunity to answer any ethics questions raised by the committee.

Please note, this is the last section of the form. When you press the 'next' button, you will submit your ethics application form. Please ensure you are ready to submit, as you can't go back and amend once you press the 'next' button. Once submitted, all forms should be downloaded using the 'Download PDF' button on the final survey response page and then sent to your supervisor for checking and signing off.

[Accessibility](#)

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